

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S8 (R,S) Exam April 2025 (2019 Scheme)

Course Code: CXT434**Course Name: SOFTWARE TESTING AND QUALITY ASSURANCE****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

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|----|--|-----|
| 1 | What are software errors and faults. | (3) |
| 2 | What are the software quality factors. | (3) |
| 3 | What is the main goal of functional testing. | (3) |
| 4 | Explain black box testing. | (3) |
| 5 | Explain the concept of total quality management (TQM). | (3) |
| 6 | List and explain the maturity levels of CMM. | (3) |
| 7 | Explain Test Reporting. | (3) |
| 8 | Explain the process of regression testing. | (3) |
| 9 | What is the scope of automation testing? | (3) |
| 10 | Define software quality metric. | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

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| 11 | a) Explain the difference between quality control and quality assurance. | (7) |
| | b) Explain the comparison between verification and validation | (7) |

OR

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| 12 | a) Explain in detail software quality and different views of software quality. | (7) |
| | b) What is testing? Explain the need of software testing in software development. | (7) |

Module II

- 13 a) Describe the classification of white box testing. (7)
b) Explain the different techniques used in black box testing. (7)

OR

- 14 a) Explain in detail the different phases of software testing. (7)
b) Explain in detail non-functional testing. (7)

Module III

- 15 a) Discuss what is six sigma quality. How is it achieved? (7)
b) Explain the relationship between quality factor and criteria. (7)

OR

- 16 a) List and explain the principles of ISO 9000:2000 standard. (7)
b) Explain in detail CMM (Capability Maturity Model). (7)

Module IV

- 17 a) Write a short note on tools for regression testing. (7)
b) What is program slicing? Explain dynamic program slicing. (7)

OR

- 18 a) Explain in detail test planning. (7)
b) Explain the Importance of regression testing in software maintenance. (7)

Module V

- 19 a) Discuss the criteria and steps in selecting the testing tool for automation. (7)
b) Explain the framework for test automation. (7)

OR

- 20 a) Explain design and architecture for automation. (7)
b) Explain the continuous improvement initiatives in software quality. (7)
